SQL Analysis Summary

E-Commerce Customer Drop-off & Conversion Analysis

SQL Server • Power BI • Python for data preparation where applicable

Purpose: Document the SQL work completed to validate the dataset, build session-level analysis tables, and prepare insights for Power BI.

Executive Summary

SQL Server was used to validate the prepared e-commerce event dataset, create session-level funnel tables, and analyze conversion patterns across overall sessions, categories, brands, price bands, and event hours. These results support three Power BI dashboards.

The clearest observed bottleneck is the transition from product view to cart addition: approximately 4.31% of view sessions progressed to a cart, while approximately 95.69% did not. This is descriptive analysis and does not prove the causes of customer behavior.

Scope note: prepared data represents October 1, 2019 event data (00:00 to 22:43), not the full month.

1. Database & Main Table

Database: **ecommerce_dropoff**
Main table: **dbo.ecommerce_events**
Prepared rows: **499,712**
Columns: **16**

The main table contains event-level records such as event type, product/category identifiers, brand, price, user ID, session ID, and date/time attributes.

2. Data Validation

Validation item	Result
Total records	499,712
Unique users	89,124
Unique sessions	117,455
View events	481,790
Cart events	8,165
Purchase events	9,757
Duplicate rows	0
Null category_code values	158,366
Null brand values	71,843

These checks established dataset size, event distribution, duplicate count, and missing-value levels before funnel analysis.

3. Session-Level Funnel Analysis

Sequential stage	Sessions	Definition
View	117,406	Sessions containing a view
Cart	5,066	View sessions that also added to cart
Purchase	2,875	View sessions with cart and purchase

Metric	Approx. value	Definition
View-to-cart conversion	4.31%	Cart sessions ÷ view sessions
View-to-cart drop-off	95.69%	100% – view-to-cart conversion
Cart-to-purchase conversion	56.77%	Cart-and-purchase sessions ÷ cart sessions
Sequential view-to-purchase	2.45%	View + cart + purchase sessions ÷ view sessions

Dashboard 1's Purchase Sessions KPI counts all sessions with any purchase (approximately 8,256, displayed as 8K). The sequential funnel purchase stage counts only sessions with view + cart + purchase (2,875, displayed as 3K). Both are valid for their definitions.

4. SQL Tables Created

SQL object	Purpose
dbo.ecommerce_events	Main event-level dataset
dbo.session_funnel	Overall session-level funnel
dbo.category_session_funnel	Category-level funnel metrics
dbo.brand_session_funnel	Brand-level funnel metrics
dbo.price_session_funnel	Price-band funnel metrics
dbo.hour_session_funnel	Hourly session-level analysis

5. Category Analysis

The category table supports comparisons of view, cart, and purchase sessions, conversion rates, and drop-off patterns. Example: electronics.smartphone had 39,204 view sessions, 3,606 cart sessions, and 2,232 sequential purchase sessions in the results reviewed.

Category totals may overlap because a session can interact with multiple categories. Missing category values also limit coverage.

6. Brand Analysis

The brand table supports comparisons for brands such as Samsung, Apple, Oppo, Huawei, and Xiaomi. This analysis is descriptive and does not prove that brand alone caused conversion differences.

7. Price-Band Analysis

Price band	Views	Carts	Purchases	View→Cart	Cart→Purchase	View→Purchase
100–250	46,724	1,920	1,120	4.11%	58.33%	2.40%
250–500	33,504	1,215	746	3.63%	61.40%	2.23%
500–1,000	23,621	841	513	3.56%	61.00%	2.17%
1,000+	12,805	419	237	3.27%	56.56%	1.85%
50–100	27,277	330	160	1.21%	48.48%	0.59%
0–50	31,283	481	177	1.54%	36.80%	0.57%

The 100–250 band had the highest view volume and highest view-to-purchase rate in this summary. The 0–50 and 50–100 bands had lower view-to-purchase rates. Product mix and customer intent may also influence these patterns.

8. Customer Behavior Analysis

Views per session	Sessions	Purchase sessions	Purchase rate
1 view	44,034	1,462	3.32%
2–3 views	34,712	3,225	9.29%
4–5 views	14,696	1,371	9.33%
6+ views	24,013	2,198	9.15%

Sessions with two or more views had higher purchase conversion than single-view sessions, but conversion did not rise steadily as view count increased.

The cart-presence comparison found 2,882 purchases among 5,077 cart sessions (56.77%) and 5,374 purchases among 112,378 sessions without a cart (4.78%). This compares any purchase in each group, not a strictly sequential funnel rate.

Purchase conversion by distinct products interacted with was 7.34% for one product, 7.25% for 2–3 products, 6.27% for 4–5 products, and 5.82% for 6+ products. Higher product variety was associated with lower conversion in this sample.

9. Business Implications & Recommendations

Prioritize view-to-cart: review product-page clarity, relevance, stock status, price transparency, delivery estimates, and Add to Cart visibility.

Reduce checkout uncertainty: make shipping fees, delivery dates, returns, and payment options easy to find.

Monitor segments: track conversion by category, price band, brand, and hour where data coverage permits.

Investigate low-conversion price bands: examine product mix and customer intent before changing pricing strategy.

Validate over more dates: repeat analysis across a longer time window before assuming patterns are stable.

10. SQL Skills Demonstrated

Data validation; aggregate queries; distinct session counting; conditional aggregation; session-level grouping; CASE-based segmentation; conversion and drop-off calculations; segment comparisons; and preparing analytical tables for Power BI.

Definitions & Limitations

Session-level metrics use unique session IDs and view/cart/purchase flags. Sequential View-to-Purchase requires all three stages. The data covers one day only; analyses are observational; category and brand values contain missing data; and grouped totals can differ because sessions can interact with multiple products or categories.

Dataset source: Kaggle — “eCommerce behavior data from multi-category store” by Michael Kechinov / Open CDP / REES46.
https://www.kaggle.com/datasets/mkechinov/ecommerce-behavior-data-from-multi-category-store